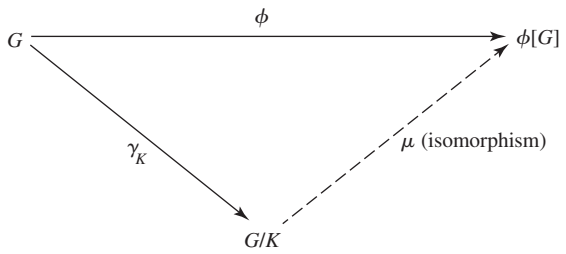
PART
IV Advanced Group Theory

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SECTION 16 ISOMORPHISM THEOREMS

There are several theorems concerning isomorphic factor groups that are known as the *isomorphism theorems* of group theory. The first of these is Theorem 12.14, which we restate for easy reference. The theorem is diagrammed in Fig. 16.1.



16.1 Figure

16.2 Theorem **(First Isomorphism Theorem)** Let $\phi : G \to G'$ be a homomorphism with kernel K, and let $\gamma_K : G \to G/K$ be the canonical homomorphism. There is a unique isomorphism $\mu : G/K \to \phi[G]$ such that $\phi(x) = \mu(\gamma_K(x))$ for each $x \in G$. ◆

The lemma that follows will be of great aid in our proof and intuitive understanding of the other two isomorphism theorems.

16.3 Lemma Let N be a normal subgroup of a group G and let $\gamma : G \to G/N$ be the canonical homomorphism. Then the map ϕ from the set of normal subgroups of G containing N to the set of normal subgroups of G/N given by $\phi(L) = \gamma[L]$ is one-to-one and onto.

Proof Theorem 13.18 shows that if L is a normal subgroup of G containing N, then $\phi(L) = \gamma[L]$ is a normal subgroup of G/N. Because $N \leq L$, for each $x \in L$ the entire coset xN in G is contained in L. Thus by Theorem 10.17, $\gamma^{-1}[\phi(L)] = L$. Consequently, if L and M are normal subgroups of G, both containing N, and if $\phi(L) = \phi(M) = H$, then $L = \gamma^{-1}[H] = M$. Therefore ϕ is one-to-one.

If H is a normal subgroup of G/N, then $\gamma^{-1}[H]$ is a normal subgroup of G by Theorem 13.18. Because $N \in H$ and $\gamma^{-1}[\{N\}] = N$, we see that $N \subseteq \gamma^{-1}[H]$. Then $\phi(\gamma^{-1}[H]) = \gamma[\gamma^{-1}[H]] = H$. This shows that ϕ is onto the set of normal subgroups of G/N. ◆

If H and N are subgroups of a group G, then we let

$$HN = \{hn \mid h \in H, n \in N\}.$$

We define the **join** $H \vee N$ of H and N as the intersection of all subgroups of G that contain HN; thus $H \vee N$ is the smallest subgroup of G containing HN. Of course $H \vee N$ is also the smallest subgroup of G containing both H and N, since any such subgroup must contain HN. In general, HN need not be a subgroup of G. However, we have the following lemma.

16.4 Lemma If N is a normal subgroup of G, and if H is any subgroup of G, then $H \vee N = HN = NH$. Furthermore, if H is also normal in G, then HN is normal in G.

Proof We show that HN is a subgroup of G, from which $H \vee N = HN$ follows at once. Let $h_1, h_2 \in H$ and $n_1, n_2 \in N$. Since N is a normal subgroup, we have $n_1 h_2 = h_2 n_3$ for some $n_3 \in N$. Then $(h_1 n_1)(h_2 n_2) = h_1(n_1 h_2)n_2 = h_1(h_2 n_3)n_2 = (h_1 h_2)(n_3 n_2) \in HN$, so HN is closed under the induced operation in G. Clearly $e = ee$ is in HN. For $h \in H$ and $n \in N$, we have $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}n_4$ for some $n_4 \in N$, since N is a normal subgroup. Thus $(hn)^{-1} \in HN$, so $HN \leq G$. A similar argument shows that NH is a subgroup, so $NH = H \vee N = HN$.

Now suppose that H is also normal in G, and let $h \in H, n \in N$, and $g \in G$. Then $ghng^{-1} = (ghg^{-1})(gng^{-1}) \in HN$, so HN is indeed normal in G. ◆

We are now ready for the second isomorphism theorem.

16.5 Theorem **(Second Isomorphism Theorem)** Let H be a subgroup of G and let N be a normal subgroup of G. Then $(HN)/N \simeq H/(H \cap N)$.

Proof Since $N \leq HN \leq G$ and N is a normal subgroup of G, N is a normal subgroup of HN, which allows us to form the group HN/N. We define a map $\phi : H \to HN/N$ by $\phi(h) = hN$. The map ϕ is a homomorphism since for any $h_1, h_2 \in H$,

$$\phi(h_1 h_2) = (h_1 h_2)N = (h_1 N)(h_2 N) = \phi(h_1)\phi(h_2).$$

The map ϕ maps onto HN/N since any element of HN/N can be written as hnN for some $h \in H$ and $n \in N$ and $hnN = hN = \phi(h)$. We now compute the kernel of ϕ.

$$\text{Ker}(\phi) = \{h \in H \mid hN = N\} = \{h \in H \mid h \in N\} = H \cap N$$

By the First Isomorphism Theorem, Theorem 16.2, the map $\mu : H/(H \cap N) \to HN/N$ defined by $\mu(h(H \cap N)) = hN$ is an isomorphism. ◆

16.6 Example Let $G = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}, H = \mathbb{Z} \times \mathbb{Z} \times \{0\}$, and $N = \{0\} \times \mathbb{Z} \times \mathbb{Z}$. Then clearly $HN = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ and $H \cap N = \{0\} \times \mathbb{Z} \times \{0\}$. We have $(HN)/N \simeq \mathbb{Z}$ and we also have $H/(H \cap N) \simeq \mathbb{Z}$. ▲

16.7 Example Let $G = \mathbb{Z}$, $N = \langle n \rangle$, and $H = \langle h \rangle$ where n and h are positive. The group $\mathbb{Z}$ is abelian, so N is a normal subgroup of Z. Since $\mathbb{Z}$ is an additive group, we write $N + H$ instead of NH to avoid confusion. The group $N + H = \langle \gcd(n, h) \rangle$ since anything in $N + H$ is a multiple of $\gcd(n, h)$ and $\gcd(n, h) = xn + yh \in N + H$ for some integers x and y. Also, $N \cap H = \langle \text{lcm}(n, h) \rangle$ since $a \in N \cap H$ if and only if a is a multiple of both n and h. The Second Isomorphism Theorem states that $(N + H)/N = \langle \gcd(n, h) \rangle / \langle n \rangle$

is isomorphic with $H/(N \cap H) = \langle h \rangle / \langle \text{lcm}(n,h) \rangle$. In Exercise 10, you will be asked to prove that if $a, b \in \mathbb{Z}^+$ and a divides b, then $|\langle a \rangle / \langle b \rangle| = b/a$. Since $\langle \gcd(n,h) \rangle / \langle n \rangle \simeq \langle h \rangle / \langle \text{lcm}(n,h) \rangle$, we have

$$\frac{n}{\gcd(n,h)} = \frac{\text{lcm}(n,h)}{h}$$
$$nh = \gcd(n,h)\text{lcm}(n,h),$$

which provides a complicated way of proving a basic number theory fact! We conclude that

$$\langle \gcd(n,h) \rangle / \langle n \rangle \simeq \langle h \rangle / \langle \text{lcm}(n,h) \rangle \simeq \mathbb{Z}_{\frac{n}{\gcd(n,h)}}$$

since a factor group of a cyclic group is cyclic. ▲

If H and K are two normal subgroups of G and $K \leq H$, then H/K is a normal subgroup of G/K. The third isomorphism theorem concerns these groups.

16.8 Theorem **(Third Isomorphism Theorem)** Let H and K be normal subgroups of a group G with $K \leq H$. Then $G/H \simeq (G/K)/(H/K)$.

Proof Since K is a subgroup of H, for any $g \in G$, $gK \subseteq gH$. That is, each left coset of K is completely contained in one coset of H. We define $\phi : G/K \to G/H$ by $\phi(gK) = gH$. That is, we map a coset of K to the coset of H that contains it. Again, our strategy is to use the First Isomorphism Theorem. The map ϕ is a homomorphism since for any $g_1, g_2 \in G$,

$$\begin{aligned}\phi((g_1K)(g_2K)) &= \phi((g_1g_2)K) = (g_1g_2)H \\ &= (g_1H)(g_2H) \\ &= \phi(g_1K)\phi(g_2K).\end{aligned}$$

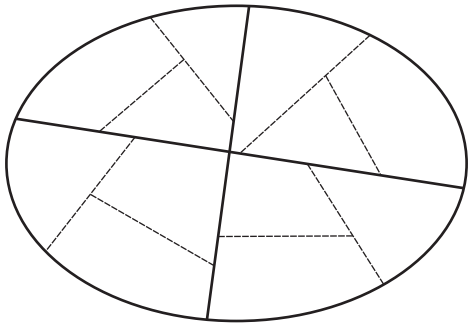
The map ϕ maps onto G/H since for any coset $gH \in G/H$, $\phi(gK) = gH$. We now compute the kernel of ϕ.

$$\begin{aligned}\text{Ker}(\phi) &= \{gK \in G/K \mid gH = H\} \\ &= \{gK \in G/K \mid g \in H\} \\ &= H/K.\end{aligned}$$

By the First Isomorphism Theorem $(G/K)/(H/K)$ is isomorphic with G/H and a formula for an isomorphism $\mu : (G/K)/(H/K) \to G/H$ is $\mu((gK)H/K) = gH$. ◆

The formula for the isomorphism in the previous proof says that if we collapse the subgroup K in G to form G/K and then collapse all the K cosets inside of H we have the same group as collapsing the subgroup H. Figure 16.9 illustrates the situation. Think of the large ellipse as being the group G. The cosets of H are the sets bounded by the thick solid lines. The cosets of K are the smaller sets inside the cosets of H. The set G/H consists of the cosets of H, which are represented by the four larger areas, each of which is one point in G/H. On the other hand, G/K is represented by the twelve smaller sets each collapsed to a point. Then $(G/K)/(H/K)$ collapses each of the three small areas in the same H coset to their common H coset. Either way, we end up collapsing each H coset to a point.

It is sometimes difficult to think about what groups look like when they contain more than one factor group in their definition. For example, what does an element of $(G/K)/(H/K)$ really look like? Keep in mind that a factor group has cosets of the subgroup as elements. Example 16.10 is intended to clarify what the group $(G/K)/(H/K)$ looks like and to explicitly show what the isomorphism of Theorem 16.8 looks like.



16.9 Figure

16.10 Example Let $G = \mathbb{Z}_8$, $H = \langle 2 \rangle = \{0, 2, 4, 6\} < G$, and $K = \langle 4 \rangle = \{0, 4\} < H$. We list the elements of each of the factor groups used in Theorem 16.8.

$$G/K = \{\{0, 4\}, \{1, 5\}, \{2, 6\}, \{3, 7\}\}$$
$$G/H = \{\{0, 2, 4, 6\}, \{1, 3, 5, 7\}\}$$
$$H/K = \{\{0, 4\}, \{2, 6\}\}$$

Before listing the elements of $(G/K)/(H/K)$ we note that any element of this group would be a coset of the subgroup $H/K < G/K$. So each element is a set whose elements are sets.

$$(G/K)/(H/K) = \{\{\{0, 4\}, \{2, 6\}\}, \{\{1, 5\}, \{3, 7\}\}\}$$

Comparing G/H and $(G/K)/(H/K)$, we see that each element in G/H is the union of two cosets of K in G/K. Also, the isomorphism in Theorem 16.8 is the map $\phi : (G/K)/(H/K) \to G/H$ defined by

$$\phi(\{\{0, 4\}, \{2, 6\}\}) = \{0, 4, 2, 6\} \quad \text{and}$$
$$\phi(\{\{1, 5\}, \{3, 7\}\}) = \{1, 5, 3, 7\}.$$

So as illustrated in Figure 16.9, collapsing H to form G/H can be accomplished by first collapsing K and then collapsing H/K. ▲

EXERCISES 16

Computations

In using the three isomorphism theorems, it is often necessary to know the actual correspondence given by the isomorphism and not just the fact that the groups are isomorphic. The first six exercises give us training for this.

1. Let $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_3$ be the homomorphism such that $\phi(1) = 2$.

- **a.** Find the kernel K of ϕ.
- **b.** List the cosets in $\mathbb{Z}_{12}/K$, showing the elements in each coset.
- **c.** Give the correspondence between $\mathbb{Z}_{12}/K$ and $\mathbb{Z}_3$ given by the map μ described in Theorem 16.2.

2. Let $\phi : \mathbb{Z}_{18} \to \mathbb{Z}_{12}$ be the homomorphism where $\phi(1) = 10$.

- **a.** Find the kernel K of ϕ.
- **b.** List the cosets in $\mathbb{Z}_{18}/K$, showing the elements in each coset.
- **c.** Find the group $\phi[\mathbb{Z}_{18}]$.
- **d.** Give the correspondence between $\mathbb{Z}_{18}/K$ and $\phi[\mathbb{Z}_{18}]$ given by the map μ described in Theorem 16.2.

3. In the group $\mathbb{Z}_{24}$, let $H = \langle 4 \rangle$ and $N = \langle 6 \rangle$.

a. List the elements in HN (which we might write $H + N$ for these additive groups) and in $H \cap N$.

b. List the cosets in HN/N, showing the elements in each coset.

c. List the cosets in $H/(H \cap N)$, showing the elements in each coset.

d. Give the correspondence between $H/(H \cap N)$ and HN/N described in the proof of Theorem 16.5.

4. Repeat Exercise 3 for the dihedral group D_6 with $N = \{\iota, \mu, \rho^2, \mu\rho^2, \rho^4, \mu\rho^4\}$ and $H = \langle \rho \rangle$.

5. In the group $G = \mathbb{Z}_{24}$, let $H = \langle 4 \rangle$ and $K = \langle 8 \rangle$.

a. List the cosets in G/H, showing the elements in each coset.

b. List the cosets in G/K, showing the elements in each coset.

c. List the cosets in H/K, showing the elements in each coset.

d. List the cosets in $(G/K)/(H/K)$, showing the elements in each coset.

e. Give the correspondence between $(G/K)/(H/K)$ and G/H described in the proof of Theorem 16.8.

6. Repeat Exercise 5 for the dihedral group $G = D_8$, $H = \langle \rho^2 \rangle = \{\iota, \rho^2, \rho^4, \rho^6\}$, and $K = \langle \rho^4 \rangle = \{\iota, \rho^4\}$.

Theory

7. Show directly from the definition of a normal subgroup that if H and N are subgroups of a group G, and N is normal in G, then $H \cap N$ is normal in H.

8. Let H, K, and L be normal subgroups of G with $H < K < L$. Let $A = G/H, B = K/H$, and $C = L/H$.

a. Show that B and C are normal subgroups of A, and $B < C$.

b. To what factor group of G is $(A/B)/(C/B)$ isomorphic?

9. Let K and L be normal subgroups of G with $K \vee L = G$, and $K \cap L = \{e\}$. Show that $G/K \simeq L$ and $G/L \simeq K$.

10. Use one of the Isomorphism Theorems to prove that if $a, b \in \mathbb{Z}^+$ and a divides b, then $|a\mathbb{Z}/b\mathbb{Z}| = b/a$.

11. Let G be a group with subgroups $H^* \leq H \leq G$ and $K^* \leq K \leq G$. Prove that the sets $H^*(H \cap K^*) \cap (H \cap K)$ and $(H^* \cap K)(H \cap K^*)$ are equal.

SECTION 17 SYLOW THEOREMS

The Fundamental Theorem for Finitely Generated Abelian Groups (Theorems 9.12 and 9.14) give us complete information about all finite abelian groups. The study of finite nonabelian groups is much more complicated. The Sylow theorems give us some important information about them.

The Theorem of Lagrange says that if H is a subgroup of a finite group G, then the order of H divides the order of G. The Fundamental Theorem for Finitely Generated Abelian Groups implies that if k divides the order of a finite abelian group G, then G has a subgroup of order k. The situation is different for nonabelian groups. Example 13.6 shows that although A_4 has 12 elements, it has no subgroup of order 6. Furthermore, for $n \geq 5$, A_n can have no subgroup of index 2 since A_n is simple and any subgroup of index 2 is a normal subgroup. On the positive side, Cauchy's theorem (14.20) says that if the prime number p divides the order of a group G, then G has a subgroup of order p. The Sylow theorems generalize Cauchy's theorem to show that if p^n divides the order of G, then G has a subgroup of order p^n as long as p is a prime number. Furthermore, Sylow's theorems gives us information about relationships between these subgroups of G. As we shall see, this information is very useful in the study of finite nonabelian groups.

Proofs of the Sylow theorems give us another application of action of a group on a set described in Section 14. This time, the set itself is formed from the group; in some instances the set is the group itself, sometimes it is a collection of cosets of a subgroup, and sometimes it is a collection of subgroups.